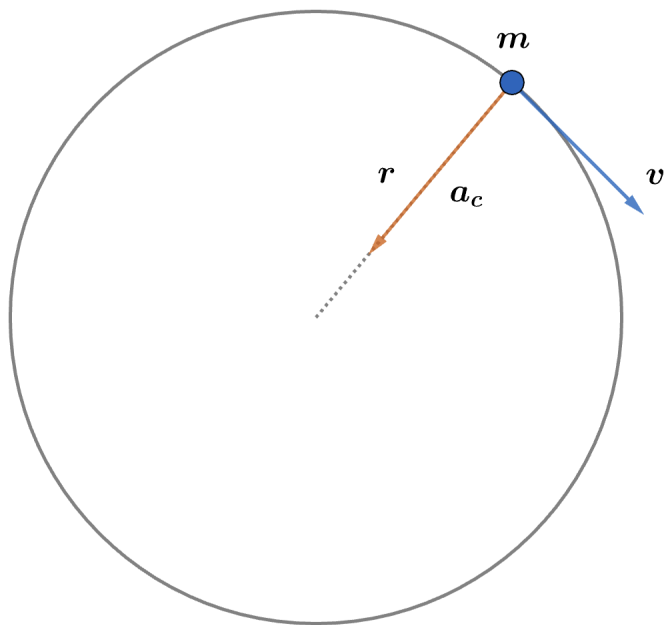


2010 F=ma Exam: Problem 7

Kevin S. Huang



Recall the centripetal acceleration is given by

$$a_c = \frac{v^2}{r}$$
$$v = \sqrt{a_c r}$$

and the angular momentum is given by

$$L = mvr = mr^{3/2}\sqrt{a_c}$$

When Harry passes out, we have $a_c = 5g$ which yields

$$L = mr^{3/2}\sqrt{5g} = (50\text{ kg})(2\text{ m})^{3/2}\sqrt{5(10\text{ m/s}^2)} = 1000\text{ kg m}^2/\text{s}$$

so the answer is D.